1.2 Why Vector Calculus

Training is an optimization problem. Linear regression is the first example.

1. Training is optimization

A model is a function of **parameters** $\boldsymbol{\theta}$. A **cost** $J(\boldsymbol{\theta})$ says how well those parameters explain the data. Training is

$$\hat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}).$$

Three pictures you will see again:

- **Linear regression.** $\boldsymbol{\theta}$ are weights. J is squared error (or a likelihood).
- **Auto-encoders** (Module 6). $\boldsymbol{\theta}$ are layer weights. J is reconstruction error. The chain rule is how you differentiate through the stack.
- **Gaussian mixtures** (Module 5). $\boldsymbol{\theta}$ are means, covariances, and mixture weights. J is a (negative) likelihood.

If you cannot write J and say what $\boldsymbol{\theta}$ is, you do not yet have a learning problem.

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2. Linear regression, written so the calculus is visible

Training pairs $(y_n, \mathbf{x}_n)$, $n=1,\dots,N$. Prediction $\hat y_n = \hat{\boldsymbol{\theta}}^\top \mathbf{x}_n$. Cost

$$J(\boldsymbol{\theta}) = \sum_{n=1}^N \bigl(y_n - \boldsymbol{\theta}^\top \mathbf{x}_n\bigr)^2.$$

Set $\nabla J = \mathbf{0}$. In matrix form that is the **normal equation**

$$(\mathbf{X}^\top\mathbf{X})\hat{\boldsymbol{\theta}} = \mathbf{X}^\top\mathbf{y}, \qquad \hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$$

when $\mathbf{X}^\top\mathbf{X}$ is invertible. The inverse existing is a linear-algebra fact (full column rank). The $\nabla J=\mathbf{0}$ step is calculus.

For two weights the surface J is a convex bowl. One local minimum, and it is global. Neural nets will not be this kind.

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3. What Module 1 still owes you

To minimize a smooth J you need:

- a **gradient** (first derivatives) — when to stop, which way is downhill;

- a **Hessian** (second derivatives) — how curved the bowl is;
- later, **constraints** (week 3) when $\boldsymbol{\theta}$ is not free in $\mathbb{R}^l$.

The next two notes are those derivatives. Week 2 turns them into algorithms.